

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years building scalable web applications using Node.js, TypeScript, and NoSQL databases (MongoDB, Redis, Firebase). Expert in architecting real-time sync systems and executing zero-downtime cloud migrations. Proven track record leading end-to-end feature development, from SPA frontend implementation to backend API design and deployment automation. Strong remote collaboration skills across distributed teams, with experience shipping production LLM integrations and performance-optimized applications serving millions of users.

CORE COMPETENCIES

Node.js / TypeScript

MongoDB / NoSQL

Real-time Sync Systems

Cloud Migrations (GCP)

SPA Development

LLM Integration

Test-Driven Development

Remote Collaboration

WORK EXPERIENCE

MTailor

Software Engineer — US Remote

Sep 2022 – Present

- Led end-to-end migration of 20+ Node.js applications from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to CTO
- Architected **real-time bidirectional sync** between MongoDB and Firestore using Node.js and Google Pub/Sub for event-driven message processing
- Built scalable 3D visualization feature using video overlay and ffmpeg, **increasing buyer conversion**
- Implemented Express Shipping feature generating **\$40 additional revenue per order**
- Reduced infrastructure costs by **\$5,000/month** through strategic AWS to GCP migration

MTailor

Contractor — US Remote

Mar 2022 – Jul 2022

- Built migration scripts to read MongoDB data and write to Firestore with complex transformation logic
- Authored technical documentation for Firebase SDKs (Node.js, Python) enabling team adoption
- Trained operations team on Firebase Dashboard workflows and backup procedures

Mind2matter.co

Software Engineer — US Remote

Jan 2022 – Jul 2022

- Built backend systems for DeFi applications using Web3, Node.js, and TypeScript
- Delivered responsive SPA UIs from Figma designs under tight deadlines using modern HTML/CSS

Dr Wealth

Software Engineer — Singapore Remote

Aug 2021 – Jan 2022

- Optimized PWA delivering **real-time stock market prices** from Firebase backend
- Built ad-hoc jobs to query 2M+ Firestore records, syncing customer prices from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js

CodeBits

Software Engineer — Uganda Remote

Jan 2020 – Jul 2021

- Led team of 4 developers** building legal tech systems for NGOs in Uganda
- Architected microservices backend using Apache Kafka, Docker, and Kubernetes for FIDA Uganda case management
- Developed cross-platform mobile app using React Native and Expo with Firebase push notifications
- Built USSD service communicating with Node.js backend via gRPC for legal aid providers

PROJECTS

Real-time Pub/Sub Sync Pipeline Production

Built live MongoDB ↔ Firestore sync pipeline supporting parallel production traffic during zero-downtime migration. Implemented event ordering and conflict resolution patterns.

Node.js, Google Pub/Sub, MongoDB, Firestore

Healthcare WhatsApp Chatbot LLM Integration

Integrated OpenAI GPT + Google Vision for prescription OCR via WhatsApp Business API. Production LLM system handling natural language queries.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Marketplace Recommendation Engine ML/Data

Built collaborative + content-based filtering engine in MongoDB for service provider matching.

Node.js, MongoDB, Python

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Languages: JavaScript, TypeScript, Python, Go, Rust **Backend:** Node.js, Express.js, Firebase Cloud Functions, gRPC

Frontend: React, React Native, HTML5, CSS3, Tailwind **Databases:** MongoDB, Firestore, Redis, SQL

Cloud: GCP (Pub/Sub, Cloud Functions), AWS (S3, EC2), Firebase **DevOps:** Docker, Kubernetes, GitHub Actions, Webpack

Other: Apache Kafka, NATS, ffmpeg, Web3